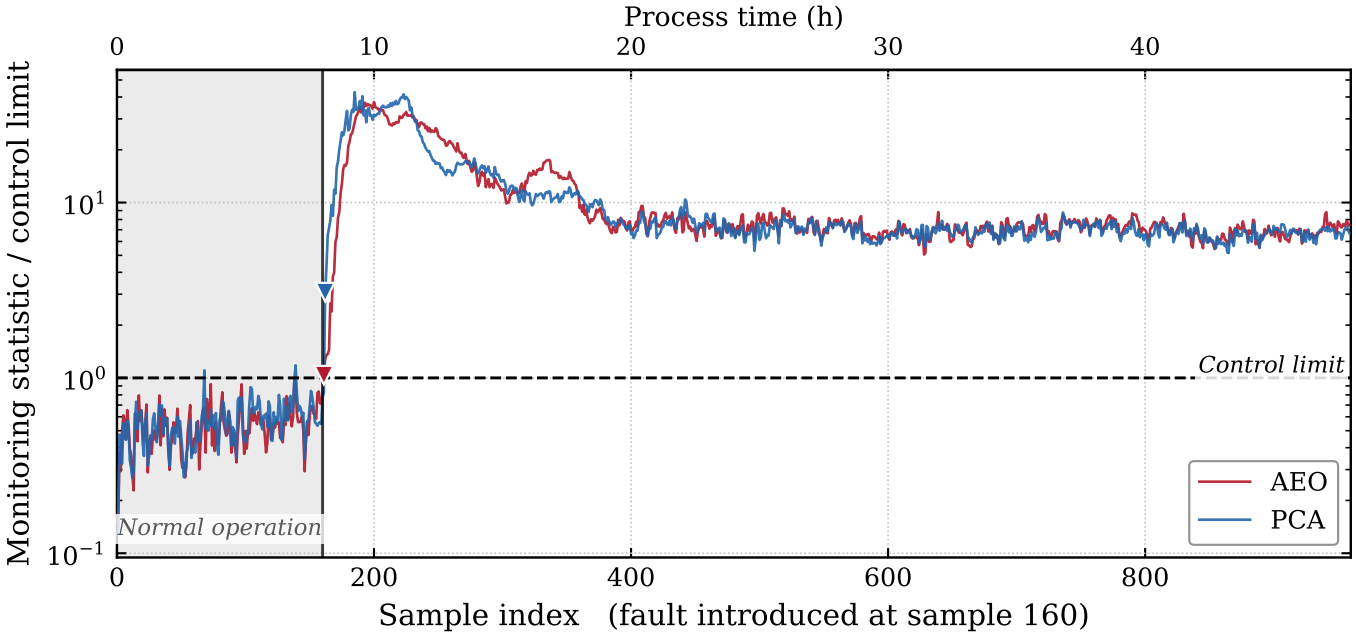
Fault Detection Performance for IDV(1)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	4.3 samples (13 min)	MDR 0.005	detected in 7/7 runs
PCA	delay	1.6 samples (5 min)	MDR 0.002	detected in 7/7 runs

PCA detects 2.7 samples (8 min) earlier